

Weakly-supervised Learning in Computer Vision and Medical Imaging

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Supervised learning

- ▶ Achieves great success, surpassing human-level error rate on ImageNet classification 2017.¹
- ▶ SENet Top 5 error rate 2.43%, human Top 5 error rate 5.1%
- ▶ Requires large amount of accurate ground truth labels.

¹Olga Russakovsky et al. Imagenet large scale visual recognition challenge. IJCV 2015

Weakly-supervised learning

Weakly-supervised learning aims to learn from imperfect scenario²:

- ▶ Incomplete supervision
Only a subset of samples labeled and large amount of unlabeled samples.
- ▶ Inaccurate supervision
Ground truth labels contain noise.
- ▶ many others ...

²Zhi-Hua Zhou. A brief introduction to weakly supervised learning. National science review, 2018

A simple demo of each scannerio

CIFAR 100



Ground Truth

Elephant

Incomplete

Inaccuarte

Bear

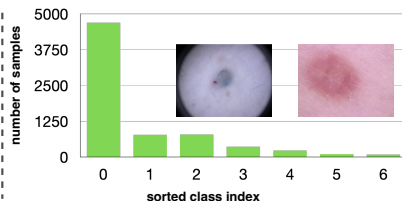
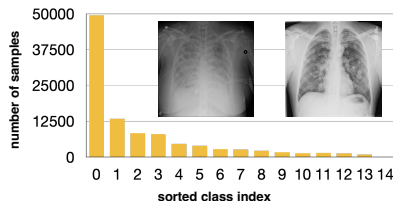
Datasets: computer vision & medical imaging analysis (MIA)

- ▶ CV: Multi-class, balance distribution, good pre-train weights (ImageNet)
- ▶ MIA: Multi-label, extreme imbalanced distribution, domain gap with ImageNet pre-train³

This impose challenges, particularly for weakly-supervised methods on MIA datasets.

³Raghu, Maithra, et al. "Transfusion: Understanding transfer learning for medical imaging." NeruIPS (2019).

A quick look at medical datasets we used



Left is NIH Chest X-ray14 (multi-label), right is ISIC2018 (multi-class)

Incomplete supervision

Two typical approaches:

- ▶ Semi-supervised Learning (SSL)

No oracle

Learning from all samples (labeled + unlabeled)
aiming for best performance.

- ▶ Active learning (AL)

With oracle

Learning from selected samples (labeled + most informative
unlabeled samples)
aiming for best performance, under a fixed budget.

We will be focusing on SSL method.

Self-supervised pre-training

Self-supervised pre-training is a great way for SSL methods

- ▶ Agnostic to multi-class/multi-label
- ▶ Robust to data imbalance ⁴
- ▶ Meaningful weight for MIA datasets

we investigate contrastive learning based self-supervised pre-training on NIH Chest X-ray14, CheXpert and ISIC2018 datasets for SSL setup.

⁴ Liu, Hong, et al. "Self-supervised learning is more robust to dataset imbalance". ICLR 2021.

Self-supervised pre-training (Cont.)

MIA adaptations:

- ▶ Batch size issue ✓
Follow MoCo⁵, we used memory bank.
- ▶ Random cropping may break in multi-label ✓
Multiple crops to estimate a Gaussian distribution and sample positive from Gaussian ⁶.
- ▶ Some augmentations are not needed ✓
Remove random grayscale, color distortion.
- ▶ MIA unique augmentations ×
Elastic transformation, histogram equalization.
No significant improve, and they are too slow

⁵He, Kaiming, et al. "Momentum contrast for unsupervised visual representation learning". CVPR, 2020.

⁶Cai, Qi, et al. "Joint contrastive learning with infinite possibilities." NeurIPS 2020

Self-supervised pre-training (Cont.)

Label Percentage	2%	5%	10%	15%	20%	100%
Graph XNet*	53.00	58.00	63.00	68.00	78.00	N/A
SRC-MT*	66.95	72.29	75.28	77.76	79.23	81.75
NoTeacher	72.60	77.04	77.61	N/A	79.49	N/A
MoCo V2	65.97	73.84	77.07	79.37	80.17	81.20
Ours	74.69	78.96	79.90	80.31	81.06	82.51

Table: Mean AUC result over the 14 disease classes of Chest X-Ray14 for different label set training percentages. * indicates the methods that use Densenet169 as backbone architecture. Rest are Densenet121.

Liu, Fengbei, et al. "Self-supervised mean teacher for semi-supervised chest x-ray classification." MICCAI-MLMI 2021.

Pseudo-labelling

Pseudo-labelling is the earliest method for SSL, works for both multi-class/multi-label⁷. But:

- ▶ Estimating threshold, particularly for each class in multi-label
- ▶ High-confidence pseudo labels are biased towards data imbalance

Improving pseudo-labeling method, address these two issues.

⁷ Rizve, Mamshad Nayeem, et al. "In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning." ICLR 2021

High-informative unlabeled samples

High-confidence pseudo label vs. High-informative unlabeled samples

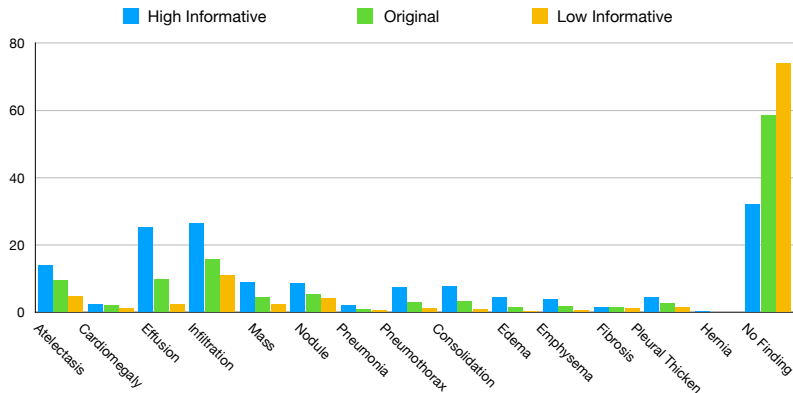
- ▶ Informativeness: feature similarity between unlabeled and labeled samples
KNN graph, Faiss library
- ▶ High-informative: unlabeled samples with low feature similarity with labeled samples
- ▶ Gaussian Mixture Model (GMM) for partitioning, no threshold
- ▶ Ensemble linear classifier and KNN classifier for supervision

Experiment results

Table: Mean AUC testing set results over the 14 disease classes of Chest X-Ray14 for different labelled set training percentages. * indicates the methods that use DenseNet-169 as backbone architecture. **Bold** number means the best result per label percentage and underline shows previous best results.

Method Type	Label Percentage	2%	5%	10%	15%	20%
Consistency based	SRC-MT*	66.95	72.29	75.28	77.76	79.23
	NoTeacher	72.60	77.04	77.61	N/A	79.49
	S ² MTS ²	<u>74.69</u>	<u>78.96</u>	<u>79.90</u>	<u>80.31</u>	<u>81.06</u>
Pseudo Label	Graph XNet*	53.00	58.00	63.00	68.00	78.00
	UPS	65.51	73.18	76.84	78.90	79.92
	Ours	74.82	79.20	80.40	81.06	81.77

Distribution visualization



Liu, Fengbei, et al. "ACPL: Anti-curriculum pseudo-labelling for semi-supervised medical image classification." CVPR 2022.

Inaccurate supervision (noisy label learning)

Chest X-ray datasets use NLP tools to extract labels from radiology report, creating noisy labels.

Three typical approaches for NLL:

- ▶ Sample selection

Small-loss assumption: model will fit clean samples first (small-loss) and fit noisy samples later (large loss)

Combine with SSL

- ▶ Robust loss

Mean Absolute Error (MAE) is robust to label noise but hard to train

Combine MAE with CE

- ▶ Transition matrix

$$p(\tilde{Y}|X) = p_{\theta_1}(\tilde{Y}|Y, X)p_{\theta_2}(Y|X)$$

Requires additional regularization, anchor points

Robust loss for MIA

For noisy label:

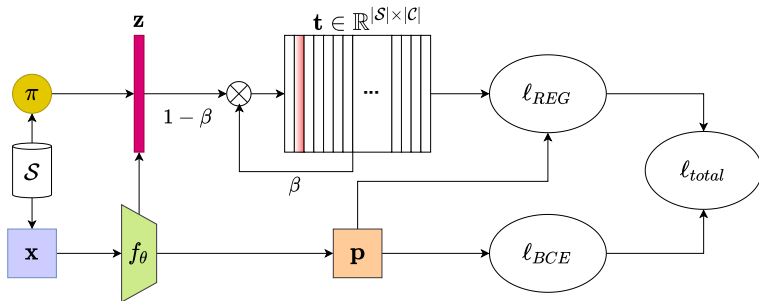
- ▶ Gradient of CE: $\nabla \ell_{CE}(\mathcal{S}, \theta) = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathbf{J}_{\mathbf{x}^i}(\theta)(\mathbf{p}^i - \mathbf{y}^i)$
 $\mathcal{S}$ is training set, $\mathbf{J}$ is Jacobian matrix, $\mathbf{p}$ is prediction and $\tilde{\mathbf{y}}$ is noisy ground truth
- ▶ Overfit noisy label because no gradient on clean label
- ▶ Maintain clean label gradient in later training
- ▶ Store a moving average of early prediction logits and regularize training
- ▶ $\ell_{REG}(\mathbf{t}^i, \mathbf{p}^i) = \log(1 - \sigma(\mathbf{t}^i)^\top \mathbf{p}^i)$

For data imbalance:

- ▶ Logits adjustment⁸: adds a label-dependent prior (the empirical class frequencies on training set) to each logits
- ▶ $\log P(Y|X) - \log P_{train}(Y) + \log P_{test}(Y)$

⁸ Menon, Aditya Krishna, et al. "Long-tail learning via logit adjustment." ICLR 2021.

Framework



Experiment results

Train on CheXpert (CXP), test on OpenI (OPI) or PadChest (PDC)

Table: Class-level and mean testing AUC on OPI and PDC for the experiment based on training on CXP. Best results for OPI/PDC are in **bold**/underlined.

Methods	CheXNet		Hermoza et al.		Ma et al.		DivideMix		Ours	
Datasets	OPI	PDC	OPI	PDC	OPI	PDC	OPI	PDC	OPI	PDC
Cardiomegaly	84.00	80.00	87.01	87.20	82.83	85.89	71.14	66.51	88.86	88.48
Edema	88.16	98.80	87.92	98.72	86.46	97.47	75.36	95.51	88.63	<u>99.60</u>
Pneumonia	65.82	58.96	65.56	53.42	61.88	54.83	57.65	40.53	64.90	<u>67.89</u>
Atelectasis	77.70	72.23	78.40	<u>75.33</u>	80.13	72.87	73.65	64.12	80.81	75.03
Pneumothorax	77.35	<u>84.75</u>	62.09	78.65	51.08	71.57	68.75	54.05	82.18	83.32
Effusion	85.81	91.84	87.00	<u>93.44</u>	88.43	92.92	78.60	79.89	83.54	89.74
Fracture	57.64	60.26	57.47	53.77	59.92	60.44	60.35	59.43	57.02	<u>62.67</u>
Mean AUC	76.64	78.12	75.06	77.29	72.96	76.57	69.36	65.72	77.99	<u>80.96</u>

Liu, Fengbei, et al. "NVUM: Non-volatile Unbiased Memory for Robust Medical Image Classification." MICCAI 2022.

Transition matrix for CV

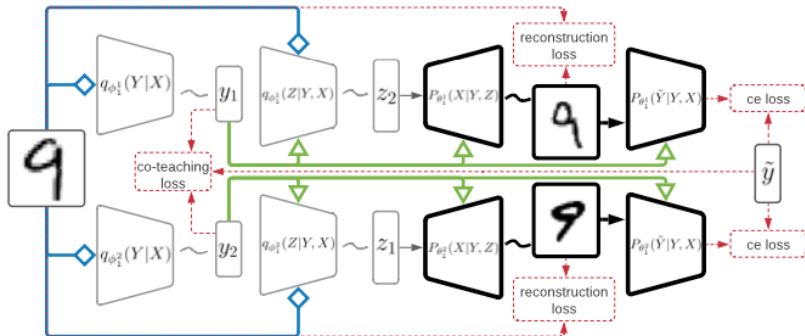
- ▶ Generative modeling solves identifiability issue of transition matrix
- ▶ $p(X, Y, \tilde{Y}, Z) = p(Y)p(Z)p(X|Y, Z)p(\tilde{Y}|Y, X)$
- ▶ Adding constraint on $p(X|Y, Z)$ will encourage identifiability of transition matrix $p(\tilde{Y}|Y, X)$

However:

- ▶ This requires a generative model (VAE, GAN)
- ▶ The ability for generating images does not help classification task⁹
- ▶ $p(Y)$ is usually assume uniform prior

⁹Dai, Zihang, et al. "Good semi-supervised learning that requires a bad gan." NeurIPS 2017.

Framework of generative modeling of transition matrix



Yao, Yu, et al. "Instance-dependent label-noise learning under a structural causal model." NeurIPS 2021.

Transition matrix for CV

Remove Z :

- ▶ $p(X, Y, \tilde{Y}) = p(Y)p(X|Y)p(\tilde{Y}|Y, X)$
- ▶ $p(X|Y)$ is intractable due to infinite number of X given only Y
- ▶ Approximate it within finite training samples:

$$p(\mathbf{x}|\mathbf{y}) \propto \frac{q(\mathbf{y}|\mathbf{x})}{\sum_{i=1}^{|\mathcal{D}|} q(\mathbf{y}|\mathbf{x}_i)}$$

$q(\mathbf{y}|\mathbf{x})$ is the variational posterior

Non-uniform $p(Y)$:

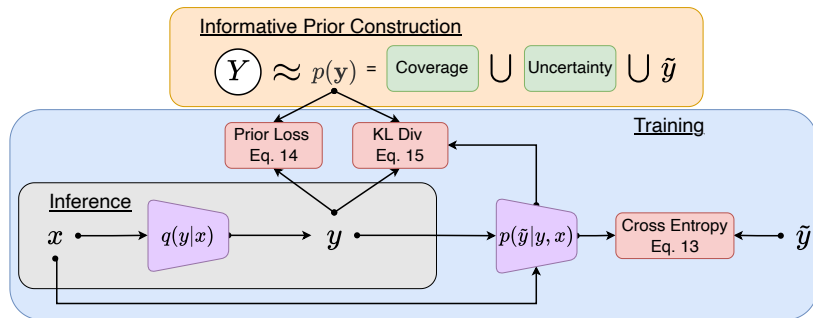
- ▶ Inspired by partial label learning decomposition¹⁰, a non-uniform $p(Y)$ is proposed:
- ▶ $p_i(\mathbf{y}(j)) = \frac{\tilde{\mathbf{y}}_i(j) + \mathbf{c}_i(j) + \mathbf{u}_i(j)}{Z}$
- ▶ $\mathbf{c}_i(j)$ is clean label coverage, $\mathbf{u}_i(j)$ is clean label uncertainty. All in one-hot form.

¹⁰Qiao, Congyu, Ning Xu, and Xin Geng. "Decompositional Generation Process for Instance-Dependent Partial Label Learning." ICLR 2022.

Our framework

Based on previous contribution, we build our framework:

► $\log p(\mathbf{x}|\mathbf{y}) = \log \frac{p(\tilde{\mathbf{y}}, \mathbf{y}, \mathbf{x})}{p(\tilde{\mathbf{y}}|\mathbf{x}, \mathbf{y})p(\mathbf{y})}.$



Experiment results

We tested under synthetic CV benchmarks, real-world CV datasets, NLP benchmarks

Method	CIFAR100			
	20%	30%	40%	50%
CE	63.94 $\pm$ 0.51	61.97 $\pm$ 1.16	58.70 $\pm$ 0.56	56.63 $\pm$ 0.69
DMI	64.72 $\pm$ 0.64	62.8 $\pm$ 1.46	60.24 $\pm$ 0.63	56.52 $\pm$ 1.18
Forward	67.23 $\pm$ 0.29	65.42 $\pm$ 0.63	62.18 $\pm$ 0.26	58.61 $\pm$ 0.44
CoTeaching	67.40 $\pm$ 0.44	64.13 $\pm$ 0.43	59.98 $\pm$ 0.28	57.48 $\pm$ 0.74
TMDNN	66.62 $\pm$ 0.85	64.72 $\pm$ 0.64	59.38 $\pm$ 0.65	55.68 $\pm$ 1.43
PartT	65.33 $\pm$ 0.59	64.56 $\pm$ 1.55	59.73 $\pm$ 0.76	56.80 $\pm$ 1.32
kMEIDTM	69.16 $\pm$ 0.16	66.76 $\pm$ 0.30	63.46 $\pm$ 0.48	59.18 $\pm$ 0.16
CausalNL	41.47 $\pm$ 0.43	40.98 $\pm$ 0.62	34.02 $\pm$ 0.95	32.13 $\pm$ 2.23
Ours	71.24$\pm$0.43	69.64$\pm$0.78	67.48$\pm$0.85	63.60$\pm$0.17

Table: Accuracy (%) on the test set for IDN problems on CIFAR100. Most results are from KMEIDTM. Top part shows discriminative and bottom shows generative models. Best results are highlighted.

Future works

- ▶ Active learning/label cleaning in MIA
- ▶ OOD samples, and its occurrence with weakly-supervised learning

Thank you!